Bank Marketing Campaign Analysis

Project Context

Ram is a Data Analyst working at a bank. The bank conducts marketing campaigns to contact customers and encourage them to subscribe to term deposits. However, the bank wants to understand which customers are more likely to subscribe and which marketing strategies are more effective.

Ram has been provided with a bank marketing dataset containing customer demographics, contact methods, campaign details, previous campaign outcomes, and subscription status. His responsibility is to analyze the data, identify customer segments with higher subscription rates, evaluate campaign performance, and develop a machine learning model to predict whether a customer is likely to subscribe to a term deposit.

Project Objectives

1. To clean and explore the dataset by identifying missing values, duplicate records, unknown values, and customer demographic patterns.

2. To analyze marketing campaign performance by examining contact methods, campaign timing, contact frequency, previous campaign outcomes, and customer subscription rates.

3. To identify potential customer segments by comparing subscription rates across different age groups, occupations, education levels, and other relevant customer characteristics.

Project Tasks

Task 1: Data Cleaning and Exploratory Data Analysis (EDA)

- Check missing values, duplicate records, and unknown values.
- Analyze customer demographics, including age, job, education, and marital status.
- Create bar charts, histograms, and count plots to understand customer distributions.
- Calculate and visualize subscription rates across different customer groups.

Task 2: Marketing Campaign Performance Analysis

- Analyze contact methods, campaign months, call duration, and number of customer contacts.
- Compare successful and unsuccessful campaign outcomes.

- Examine the relationship between previous campaign results and current subscriptions.
- Identify customer segments and campaign patterns associated with higher subscription rates.